

Week 1 Progress Report: EmbeddingGemma Powered RAG Assistant

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ABSTRACT

This report summarizes progress during Week 1 of the EmbeddingGemma Powered RAG Assistant project. The focus was on establishing the core data pipeline for ingestion, text extraction, chunking, and embedding generation—laying the foundation for retrieval and subsequent integration with large language models.

KEYWORDS

Retrieval-Augmented Generation, EmbeddingGemma, FAISS, arXiv, NLP, Information Retrieval

1 INTRODUCTION

The EmbeddingGemma Powered RAG Assistant aims to develop a retrieval-augmented generation (RAG) system for efficient scientific document search and summarization. Week 1 concentrated on constructing the end-to-end data processing pipeline that enables document ingestion, text normalization, chunking, and embedding creation, aligning with the proposed project schedule [1].

2 OBJECTIVES

The Week 1 objective, as outlined in the project proposal, was to implement the data ingestion and embedding pipeline:

“Set up ingestion, chunking, and embedding pipeline.”
[1]

3 WORK COMPLETED

The following components were designed and implemented:

- **arXiv API Client:** Automated search and download of machine learning papers.
- **LaTeX Parser:** Cleaned and extracted readable text from LaTeX sources.
- **Text Chunker:** Split text into overlapping segments of 200 tokens with 50-token overlap.
- **Embedding Modules:** Implemented both TF-IDF and transformer-based embeddings (EmbeddingGemma / MiniLM).
- **FAISS Vector Store:** Built a metadata-backed cosine-similarity search index.
- **RAG Pipeline:** Coordinated ingestion through retrieval for testing.

Verification through demo scripts confirmed correct extraction, embedding, and retrieval across sample arXiv papers.

4 RESULTS

The full pipeline successfully:

- Processed 2 arXiv papers, generating 20–100 text chunks per paper.
- Created embeddings of dimensions ~384 (transformer) and 1000–3000 (TF-IDF).
- Retrieved relevant chunks with accurate similarity scoring.
- Executed reliably on local hardware with GPU acceleration for transformer embeddings.

5 CHALLENGES

Key challenges encountered included:

- Complex LaTeX macros requiring extensive cleaning.
- Initial embedding dimension mismatches during FAISS indexing.
- GPU memory optimization for transformer batch inference.

Each issue was resolved through iterative testing and optimization.

6 NEXT STEPS

Planned Week 2 tasks:

- Integrate the Qwen-3 LLM for retrieval-augmented response generation.
- Implement citation tracking and context formatting.
- Add evaluation metrics for precision, recall, and latency.
- Begin development of a lightweight user interface demo.

7 SUMMARY

The Week 1 milestone was achieved successfully, producing a complete ingestion-to-retrieval backend for the RAG assistant. Subsequent efforts will focus on integrating generation models and performance evaluation.

REFERENCES

- (1) Justin Kobza, John Ludeke, Daniel Vlassov, and Amber Wilt. 2025. *EmbeddingGemma Powered RAG Assistant Project Proposal*. University of Illinois Urbana-Champaign.

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